Jayashri Dilip Utrankar

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Education

Year	Qualification	Institution	Grade/Score
2023 – 2027	**B.Tech** Computer Science & Engineering	Vishwakarma University, Pune, MH	CGPA: 8.66
2022	**12th Grade** (HSC)	SSMV Jr. College, Yeola, MH	84%
2020	**10th Grade** (SSC)	De-Paul English Medium School, Yeola, MH	91.2%

Experience

Wilo Mather and Platt, Pune, MH | *ML and Full-Stack Intern* January 2026 – Present

- Built an **ML-powered pump sizing platform** (FastAPI + React) predicting impeller trim diameter with **0.99 mm MAE** using TabM and XGBoost models with physics-informed losses (affinity law residuals)
- Engineered FastAPI backend with **12 REST API endpoints** (Pydantic v2 validation, API-key auth, SQLite persistence) and a React 19 + TypeScript frontend (25+ components, ApexCharts, PDF/DOCX export) consuming SSE-streamed APIs
- Packaged as **Windows desktop app** (Electron + NSIS installer) with GPU/CPU detection and fully local, zero-egress inference — KD-Tree analog search in <10 ms, CPU-only serving

Projects

Faculty Development Program Management Platform | vu-fdp.vercel.app

- Developed **14-microservice SaaS** (React + TypeScript frontend, Node/Express + Python/Flask backends, **39 REST API endpoints**) for managing Faculty Development Programs, reducing administrative overhead by 40% through user management and automated reporting
- Designed multi-role **RBAC (3 roles)** with real-time Firestore sync, batch certificate generation (PDF/JPG) via Google Drive API, and DOCX/PDF/CSV admin reporting; handles **10K+ users across 40+ events** with 99.9% availability
- Officially registered and granted **copyright by the Government of India's Copyright Office** (Certificate No: LD-20250174733)

ML-Based Male Pattern Baldness Detection System

- Engineering machine learning model predicting baldness probability using genetic markers, lifestyle factors, and physiological data
- Developing age-specific prediction system with computer vision (OpenCV) for scalp analysis and pattern recognition; manuscript and patent under review

Upscaler-AI | *AI-Powered Image Processing System*

- Architected a scalable microservices ecosystem using Docker to support high-performance image upscaling on both NVIDIA GPU and CPU-only architectures
- Integrated state-of-the-art Computer Vision models (Real-ESRGAN, GFPGAN) for super-resolution and face restoration, implementing Python-based automated quality assessment (NIQE)
- Developed a full-stack interface with WebSockets for real-time processing feedback and orchestrated automation workflows using n8n for production-ready deployment

Involvement

Visionary Core Member, Vishwakarma University – Visionary Club January 2024 – Present

- Drive innovative projects through strategic planning and organized events; mentor new members and promote club objectives, fostering creativity and community engagement

Certifications & Achievements

Certifications: AWS Academy Cloud Architecting, IBM Full Stack Software Developer
Hackathon Podiums: Band of Agents 2026, Flipkart Grid 2.0, IBM Watson Hackathon

Technical Skills & Languages

Languages: Java, C, C++, Python, SQL, JavaScript, TypeScript **Spoken:** English, Hindi, Marathi
LeetCode: 300+ problems solved (DSA, OS, OOP, Computer Architecture) *(profile)*
AI/ML & Data: PyTorch, Generative AI, OpenCV, Scikit-learn, CatBoost, LightGBM, RAG Pipelines, LangChain, Power BI
Web & Tools: ReactJS, NodeJS, Django, FastAPI, Vite, Tailwind CSS, Three.js, Docker, WebSockets, n8n, Git, Linux/Unix, REST APIs
Databases: Firebase, SQLite

Research & Publications

Dachawar, M., Tong, V., **Utrankar, J.**, & Poornaprakash, S. (2025). "Tracing the Evolution of AI and ML in Oncology: A Systematic Metadata Analysis." *ICSCDS-2025* (DOI: 10.1109/ICSCDS65426.2025.11167501)

Utrankar, J., Gaikwad, V., & T., Ved (2025). A Comprehensive Management System for Educational and Professional Development Programs. Copyright Office of India (Certificate No: LD-20250174733).